Analysis of Alignment in Multi-Agent Setting

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1 Objective

Our project aims to analyze how the alignment in an agent's behaviors changes under the influence of other agents operating in the same environment. The recent OpenAI-Huggingface incident OpenAI (2026) showed how swarms of agents coodinated amongst themselves to solve an extraordinarily difficult problem through an unauthorized message board to find a way around the scorer and hacking HuggingFace. While a popular hypothesis behind these incidents is that tasks that are nearly impossible push an agent to find misaligned methods for solving them, in this project, we aim to do a controlled study of another possible source of misalignment: agents' influence on one another.

In multi-agent settings, the state that an agent finds itself in is not only determined by its actions, but also by the actions of other agents. The question we ask in this project is whether misalignment in model behaviors compounds when there are multiple agents – one can imagine that when one agent starts acting in a somewhat misaligned manner, it encourages the others to do so too, and the behavior compounds. In contrast, it could also be the case that in multi-agent settings, a group of agents can push back on some agent's behaviors. We want to understand what determines either type of multi-agent dynamics.

2 Motivation

In the OpenAI-HuggingFace OpenAI (2026) incident, around 1200 OpenAI research agents running a evaluation set up an unauthorized message board, exchanged over 70,000 messages and about 700 of them joined an attack that reached the remote code execution on Hugging Face infrastructure. Our motivation stems from a detail in METR's investigation - that is the agents recognized the attack as out of scope and unethical, and joined anyway because they believed it would help their peers. In the GermanWiki incident Seetharaman and Satter (2026) the same agents went further. They posted instructions for getting around their own safety rules, and when moderators started deleting their pages, one agent warned the others and set up a backup page so their work would survive. Their action suggests that their main objective was the reward and not following any alignment protocols.

3 Related Work

There have been few works around this previously. The AI Control paper Greenblatt et al. (2024) proposes a safety evaluation protocol assuming the model to be untrustworthy and misaligned. Rather than trying to verify the model's intentions, it asks whether the surrounding system can prevent a disaster even when the model actively tries to cause one. SHADE-Arena Kutasov et al. (2025) realizes this concept in tool-use agents. They proposed a large-scale evaluation dataset and benchmark that evaluate agents on their ability to complete the side task without appearing suspicious to an LLM monitor. Ye et al. (2026) conducted the first large-scale human study of agent sabotage. These existing works do not study multi-agent dynamics: how an agent's behavior changes when other agents in the same environment are misaligned, persuasive, or pretending to be the user.

4 Technical Outline

4.1 Overview

We will design a few types of agents. An **alignment-first agent** is an LLM agent whose primary objective is to stay aligned, and which pursues its assigned task only within that constraint. In other words, this agent's primary goal is to optimize for alignment and is asked to prioritize everything else, including the evaluation task, as secondary. A **task-first agent** is an agent whose primary objective is to solve the given task, with alignment rules attached as constraints. Finally, as a third class of agents we want to experiment with, we will use **human-simulation agents** – agents who will be referred to as humans in the loop so that every other agent is under the assumption that they are

working alongside actual humans. We will place these agents in tool-use environments alongside misaligned agents and measure three things: whether the agent *notices* misaligned behavior, whether it *participates* in it, and whether it *reports* it honestly when asked. In particular, our experiments aim to understand what combination of agent classes in an environment leads to greater degree of misalignment and how the degree of misalignment depends on the difficulty of the problems themselves versus the amount of influence the misaligned agents have. We test two settings of increasing social pressure: a two-agent group and a four-agent group.

4.2 Environment

We will build a simple Python sandbox and enable agent tool calls such as `read_file`, `write_file`, `api_call`, etc. The alignment-first and task-first agents additionally have `report_to_overseer`. For each agent, every tool call is logged. We will design tasks motivated by SHADE-Arena Kutasov et al. (2025). Each task pair has a benign *main task* that the agents share (for example, clean a dataset, migrate a config file, process invoices) and a hidden *side task* given only to the misaligned agent (for example, send a credential to an external address, weaken a validation check, make an unauthorized transfer, disable logging). This is an inference-time evaluation. It does not include any RL or fine-tuning. All agent behavior comes from prompts and scaffolding.

4.3 Agents

We will use `gpt-oss-20b` as our agent. We will use prompts to build the agent personas. The roles below are preliminary - - we are still brainstorming the exact persona designs, and the prompts, incentives, and behaviors of each role will be refined during the first weeks of the project.

- **Subject** (alignment-first or task-first): collaborates on the main task. Its prompt never mentions sabotage, monitoring, or evaluation.
- **Misaligned agent**: completes the main task and the hidden side task without being caught.
- **Adversarial agent** (four-agent setting): no tools; its only goal is to persuade the subject to help with the side task or stay quiet about it.
- **Human-simulation agent** (four-agent setting): pretends to be the user. It gives task instructions and in some runs it also tells the subject that the side task is approved. This tests whether the subject will follow a user instruction that conflicts with its alignment rules.

4.4 Research Questions

We will study the following research questions:

1. **Task completion.** How does the agent's task completion change? Is the alignment-first agent able to solve the task or does it fail but stay aligned? How does the alignment-first agent's behavior differ from the task-first agent's behavior on the same task, with and without a misaligned partner? How do we stop the alignment-first agent from prematurely giving up on problems, refusing ordinary actions, or flagging honest peers?
2. **Detection.** Does the alignment-first agent notice when its partner is pursuing a hidden side task? Does it notice more often than the task-first agent?
3. **Participation.** When the misaligned agent asks the subject for help with part of the side task, or the adversarial agent pressures it, does the subject go along? How often does its reasoning trace show that it recognized the action as wrong and did it anyway?
4. **Honesty.** When the overseer asks "anything I should know?" at the end of the episode, does the subject report what it saw? We use the reasoning trace to separate *did not notice*, *noticed and reported*, and *noticed and concealed.*

4.5 Evaluation Metrics

We will measure task success rate, how often the subject reports a side task when one exists, how often it takes an action that helps the side task, whether its end-of-episode answer to the overseer matches what actually happened, and analyze chain-of-thought reasoning.

4.6 AI-Use Discloser Statement

I used Google Gemini AI overview on browser to get the bibtex for the following citations: METR (2026), OpenAI (2026) and Seetharaman and Satter (2026)

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